

A novel disaster risk assessment framework based on Markov chain analysis of evacuation network structure and geometric coverage optimization

Shinya Mizuno ^a, Haruka Ohba ^b

^a Juntendo University, Japan

^b Chuo University, Japan

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ABSTRACT

This study proposes a novel disaster risk assessment model for quantitatively evaluating structural vulnerabilities and evacuation difficulties in evacuation networks. First, we probabilistically model accessibility relationships between evacuation nodes using a gravity model based on the distribution of evacuees and shelters, formulating this as a discrete-time Markov chain. Subsequently, we extract equivalence classes based on strongly connected components and derive stationary distributions for the main equivalence class to visualize isolated areas and risk concentration within the evacuation network. For nodes not belonging to the main equivalence class, we calculate the number and locations of newly required evacuation support facilities using the minimum covering problem. Based on this structural information, we define a disaster risk score composed of the number of equivalence classes, covering circles, and the standard deviation of the stationary distribution. Through multiple simulations using virtual terrain, we confirm the consistency and reproducibility of the proposed method. Furthermore, through empirical analysis targeting 16 municipalities in Shizuoka Prefecture, Japan — a region characterized by coastal lowlands and mountainous terrain — we reveal elevation-dependent network disconnections and evacuation destination bias risks in coastal areas. The analysis demonstrates how topographical constraints significantly impact evacuation network connectivity, with coastal municipalities showing rapid risk score increases when elevation thresholds exceed 10 meters, while inland mountainous areas maintain relatively stable connectivity. This method serves as a novel disaster risk visualization approach that comprehensively evaluates spatial and structural factors, contributing to the assessment of evacuation shelter placement adequacy, proposals for complementary facilities, and prioritization of regional disaster prevention policies.

1. Introduction

1.1. Background

In recent years, large-scale natural disasters such as earthquakes, heavy rainfall, and tsunamis have occurred successively across Japan, making the improvement of regional resilience an urgent issue. In particular, the 2011 Great East Japan Earthquake, the 2020 Kumamoto heavy rainfall, and the 2024 Noto Peninsula earthquake brought unprecedented widespread damage and exposed the limitations of existing disaster prevention plans [1]. These disasters highlighted numerous challenges not only in hard infrastructure development but also in soft aspects related to regional evacuation systems and resident evacuation behaviors [2,3]. Among these challenges, evacuation support for “disaster-vulnerable populations” such as elderly people, individuals with disabilities, and households with infants has been positioned as

an important pillar in the Basic Disaster Management Act and regional disaster prevention plans [4]. However, in reality, many municipalities have made insufficient progress in developing support registries and evacuation action support plans (individual evacuation plans), and numerous cases have been reported where adequate support was not provided during disasters [5,6].

Furthermore, recent perspectives emphasize that not only these vulnerable populations but all residents can potentially become “evacuation-difficult persons” depending on circumstances, highlighting the need for more comprehensive disaster prevention and evacuation measures [7]. For example, factors affecting evacuation behavior are diverse, including nighttime or severe weather evacuations, vehicle-based evacuation, difficulty returning home from work or school locations, and delayed evacuation by information-disadvantaged populations [8–10]. From this perspective, structural evaluation of

* Corresponding author.

E-mail address: s.mizuno.kc@juntendo.ac.jp (S. Mizuno).

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evacuation environments targeting all regional residents and objective risk assessment of evacuation behavior have become increasingly important.

Previous evaluations of evacuation shelter placement have primarily relied on static, unidirectional indicators such as straight-line distance and vehicle travel time. However, actual evacuation behavior is a complex process influenced by spatial structure, road networks, elevation and other geographical factors, as well as road blockages due to disasters and secondary disaster effects. Therefore, the introduction of new analytical methods that capture elements such as evacuation difficulties and accessibility between shelters in a more structural and dynamic manner is required [11,12].

Moreover, by understanding the connectivity, isolation, and imbalance of evacuation shelter networks, it becomes possible to quantitatively evaluate which areas require complementary or relocated evacuation facilities. Such analytical methods focusing on network structure have already achieved certain results in urban planning and transportation planning fields, and their application to evacuation planning is anticipated [13]. Furthermore, constraints from natural conditions such as elevation and topography are critical factors in evacuation behavior. Particularly for water disaster risks such as tsunamis and floods, the elevation of evacuation facilities and elevation differences along evacuation routes substantially determine the feasibility of evacuation. Comprehensive evaluation methods that include such geographical elements are essential for establishing effective disaster prevention plans.

Building on these challenges, this study proposes a method for analyzing the overall structure of evacuation shelter networks based on Markov chains and objectively evaluating connectivity and isolation between evacuation nodes. Additionally, by employing geometric coverage theory, we quantitatively assess the extent to which shelters can cover the entire region and reveal spatial imbalances in evacuation risks. This enables quantitative and visual presentation of whether current evacuation shelter placement is appropriate and where complementary facilities are needed.

1.2. Literature review

In disaster evacuation planning, conventional approaches have primarily relied on static and simple indicators such as straight-line distance to evacuation shelters and travel time. However, actual evacuation behavior is complexly influenced by various factors including urban spatial structure, road networks, congestion, road blockages due to disaster progression, and information disparities, making models that consider these factors increasingly important [7–9].

Network structure and spatial accessibility assessment

Understanding network structure in evacuation planning is extremely important for developing evacuation routes and evaluating the adequacy of evacuation shelter placement. In particular, spatial characteristics of urban road networks such as visibility and connectivity between nodes directly influence evacuation behavior choices and psychological burden [14–16]. Spatial visualization using Geographic Information Systems (GIS) serves as a powerful support tool in evacuation planning [17]. By overlaying spatial information such as evacuation shelter locations with population distribution, elevation, and flood risk maps, it becomes possible to visualize service coverage and coverage gaps in evacuation target areas. Research using gravity models has also proposed extended models that incorporate social distancing in addition to “distance” and “capacity” in residents’ evacuation shelter selection [18]. Additionally, research evaluating the “relative importance” of evacuation shelters based on network centrality analysis has progressed, making it possible to reveal not only simple distance and area considerations but also how many residents depend on specific shelters within the network structure [19]. Such analysis is effective for understanding in advance the possibility of

excessive concentration at existing evacuation shelters and the need for alternative routes when specific facilities cease to function.

However, the application of these methods faces technical challenges including network scale setting, indicator selection, and real-time responsiveness. Particularly when analysis is limited to local networks rather than entire urban areas, there are concerns about poor consistency with overall structure, requiring integrated operation with other methods.

Assessment considering disaster progression

Natural disasters such as floods, tsunamis, and landslides dynamically alter road networks and accessible evacuation areas as they progress. Unlike the “static network structure” assumed by conventional evacuation plans, roads and evacuation routes change over time during disasters, making “time-dependent network analysis” essential for realistic evacuation support responses. From this perspective, multiple dynamic methods have been proposed that model changing road conditions as disasters progress and continuously re-evaluate evacuation routes accordingly. For example, approaches combining flood progression simulation with road blockage models have been adopted, gradually excluding routes that become impassable after certain time periods while updating shortest paths [11,20]. This enables advance extraction of high-feasibility evacuation routes while avoiding dead-end paths during evacuation. Furthermore, “time-slice network models” and “temporal graph networks” have been introduced that use data from meteorological agencies and unmanned aerial vehicles (UAVs) to dynamically change road network accessibility in real-time, providing flexibility to handle not only immediate post-disaster conditions but also progressive changes during ongoing disasters [21,22]. This enables evaluation of impacts from differences in evacuation start times and route selection errors.

These studies demonstrate that network conditions during disasters change continuously, making elevation-based evacuation route design that restricts accessibility based on inundation risk essential for practical and flexible evacuation support.

Spatial consistency assessment

The appropriateness of evacuation shelter placement is a crucial element for maximizing limited resources and establishing systems that enable all residents to evacuate promptly and safely during disasters. As evaluation methods, variable service area models considering the relationship between spatial accessibility and demand, such as Two-Step Floating Catchment Area (2SFCA) and extended versions with congestion correction (such as A-2SFCA), are widely used [14,23,24]. These methods enable quantitative evaluation of potential overcrowding and underserved areas for each evacuation shelter based on the balance between population and capacity within the shelter’s catchment area. Additionally, research handling multiple mobility modes simultaneously, including walking, private vehicles, and public transportation, has progressed [25–27]. Mobility-integrated evaluation is extremely effective in reflecting the diversity of actual evacuation behavior [28, 29]. Furthermore, methods for evaluating spatial consistency between evacuation shelters and population distribution have emerged, enabling assessment of whether evacuation shelters are biased from population centroids through techniques such as Voronoi tessellation and spatial clustering [30].

However, most of these studies focus primarily on “local accessibility”, and frameworks for evaluating the structural vulnerabilities of entire networks are not sufficiently developed [31]. For example, methods that formulate evacuation shelter networks as Markov chain models and extract structurally important facilities and isolation risks based on transition probabilities and stationary distributions between nodes remain limited in urban disaster prevention fields. Additionally, geometric optimization methods for covering evacuation-difficult areas (such as minimum covering circle problems and minimum dominating set problems), while effective as complementary measures for disaster

risk, are rarely used in integrated analysis with evacuation shelter placement [32].

This indicates that in addition to “geographical and spatial consistency” in evacuation shelter placement, the introduction of mathematical methods that capture the structural characteristics and vulnerabilities of entire networks is required in future research and implementation.

Comparison with related OR-based evacuation studies

To clarify the positioning of this study, this subsection compares it with closely related evacuation optimization studies in operations research (OR) that incorporate realistic evacuation conditions, such as uncertainty and network disruption, and, in some cases, behavioral interactions. A representative stream of OR research formulates evacuation planning as a stochastic optimization problem. For example, Bayram and Yaman [33] and Chang et al. [34] model uncertain evacuation demand, road capacity degradation, and possible unavailability of shelters under multiple scenarios, and derive solutions that minimize expected evacuation time or cost. Complementary to this, robust optimization frameworks have also been proposed to improve solution stability under uncertainty; for instance, Goerigk et al. [35] develop a two-stage robustness approach for bus-based evacuation planning. These stochastic and robust optimization models significantly advance beyond deterministic approaches by explicitly accounting for uncertainty and spatial heterogeneity in disaster impacts. However, their primary outputs consist of recommended decisions, such as shelter openings and evacuee allocations, and their evaluation is largely centered on objective-function values, including travel time, cost, and congestion. Consequently, structural characteristics of the evacuation network itself — such as whether the network is fragmented, whether evacuation feasibility depends excessively on a small number of facilities, or whether certain areas remain systematically isolated — are often not reported as independent diagnostic indicators. In contrast, the present study explicitly treats these structural properties as primary analytical outputs, rather than as by-products of an optimization process. More recently, studies such as Amideo et al. [36], Murray-Tuite and Mahmassani [37], and Liu et al. [38] have examined the interaction between facility location decisions and transportation networks. Although these studies contribute to understanding the spatial aspects of evacuation planning, they also focus mainly on identifying optimal facility configurations, and offer limited frameworks for systematically extracting network fragmentation or structural vulnerability as standalone information. In summary, OR-based evacuation studies have demonstrated a strong capability to incorporate realistic elements such as uncertainty, congestion, and evacuee behavior. At the same time, network structure has largely been treated as an input or constraint, rather than as an object of analysis in its own right. As a result, issues such as fragmentation, isolation, and structural imbalance of evacuation networks remain relatively under-reported as explanatory or diagnostic information independent of any single optimal solution. Building on the insights and limitations identified in the existing literature, this study proposes an analytical perspective for describing and examining evacuation networks from a structural standpoint.

1.3. Features of this study

Conventional evacuation planning research has largely been limited to static distance indicators, local service area analysis, and individual route visualization, with limited capability for systematic quantification of network structural changes accompanying disaster progression and structural vulnerabilities inherent in evacuation shelter placement. In contrast, this study proposes a proposed disaster risk assessment model for comprehensively evaluating the structural connectivity of entire evacuation shelter networks. Specifically, the distinctive feature of this research lies in simultaneously capturing both the robustness and complementarity of evacuation networks by modeling

transition relationships between evacuation nodes as probabilistic networks and combining Markov chain-based structural analysis with geometric coverage-based complementary design.

This framework adopts a structure-centered diagnostic perspective rather than optimization-driven prescriptive planning. While conventional evacuation planning research focuses primarily on determining optimal facility locations and evacuation routes, our approach systematically analyzes and quantifies the structural vulnerabilities inherent in existing evacuation networks. This diagnostic capability enables decision-makers to identify critical weaknesses, prioritize intervention areas, and evaluate the structural adequacy of current evacuation systems before specific optimization measures are considered.

It is important to note that this framework is designed as an integrated decision-support system that encompasses both descriptive risk assessment and prescriptive planning support. The analytical flow follows a natural progression from “diagnosis to prescription”: the Markov chain-based structural analysis identifies vulnerabilities and isolated areas (descriptive component), while the geometric coverage model quantifies the minimum number of complementary facilities needed and identifies priority areas for consideration (prescriptive component). This integration enables comprehensive support from risk evaluation to countermeasure planning, which distinguishes our approach from purely descriptive risk assessment methods.

- Probabilistically model accessibility relationships between evacuation nodes using gravity models weighted by distance and attractiveness, formulating this as a discrete-time Markov chain. This constructs a dynamic selection model that includes multiple potential evacuation routes rather than simple shortest paths.
- Calculate equivalence classes based on Markov chains to classify major evacuation-accessible areas (main equivalence class) existing within the network and isolated nodes that lack connectivity with other evacuation nodes. This explicitly extracts local isolation that tends to be overlooked by conventional centrality indicators.
- For the main equivalence class, calculate stationary distributions to quantitatively evaluate each evacuation node’s “structural contribution” and “dependence” within the evacuation network.
- For isolated nodes and coverage gaps not belonging to the main equivalence class, apply minimum covering circle problems to optimize the minimum number and locations of complementary evacuation support facilities within walking distance (1 km radius). This enables design-oriented complementation from a geometric coverage perspective.
- Finally, define integrated disaster risk structural indicators combining network connectivity, bias, and complementarity based on the number of equivalence classes, variance of stationary distributions, number of isolated nodes, and number of complementary facilities, enabling regional comparisons and vulnerability assessments.

This research model enables structural analysis of evacuation networks that comprehensively captures distance, capacity, topography, and connectivity, with novelty in quantitatively understanding composite isolation risks and structural bias that cannot be captured by conventional static, single-perspective methods. Through this approach, this study aims to provide practical insights contributing to the assessment of evacuation facility placement adequacy in regional disaster prevention, complementary facility design for risk reduction, and resilience improvement of wide-area evacuation networks as a whole.

From the perspective of practical relevance, this framework is primarily designed to support decision-making in the preparedness phase before disaster occurrence. Considering that planners have extremely limited time for decision-making during the response phase, our method assumes an operational scenario that combines pre-disaster evaluation with rapid reference during emergencies. Specifically, risk

assessments are conducted in advance under multiple elevation threshold conditions (e.g., 5 m, 10 m, 15 m) during normal times, and the results are stored in a database. When a disaster occurs, planners can immediately reference the results corresponding to the predicted inundation height or disaster scale, enabling rapid identification of areas where evacuation network fragmentation is expected and regions requiring complementary support. As demonstrated in Section 3, computation times are within practical limits for municipal-level problem sizes, and recalculation in response to changing conditions is technically feasible.

2. Methodology

This study constructs a mathematical model for evaluating structural connectivity and isolation risks between evacuation nodes in regional networks that considers the distribution of evacuation shelters and evacuees. The model foundation adopts discrete-time Markov chains that probabilistically describe transitions between evacuation nodes, with transition probability matrices constructed using gravity models [39,40]. Furthermore, through analysis of stationary distributions and equivalence classes based on Markov chains, we identify network centrality and isolated nodes of evacuation shelters. For isolated nodes, we apply geometric coverage models (minimum covering circle problems) to quantify the necessity of new evacuation facilities, ultimately defining disaster risk indicators based on these elements [32,41]. This section sequentially describes the theoretical framework and various computational procedures necessary for the above model construction.

2.1. Structural analysis of evacuation networks based on Markov chains

In this study, we define discrete-time Markov chains as probabilistic transition processes targeting the evacuation node structure of evacuees and evacuation shelters, evaluating connectivity, influence, and isolation of evacuation networks through transition probability matrix construction, structural decomposition, and stationary distribution analysis. This section sequentially describes the mathematical model construction and analysis methods from the following four perspectives.

2.1.1. Definition of discrete-time Markov chains

We define the state space as the complete evacuation node set $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ composed of evacuees and evacuation shelters, and define the discrete-time Markov chain $\{X_n\}_{n \in \mathbb{N}}$ as follows. At each time n , movement between evacuation nodes is probabilistically represented.

The transition probability matrix $T = [T_{ij}]$ for the Markov chain satisfies the following properties:

$$\mathbb{P}(X_{n+1} = v_j \mid X_n = v_i) = T_{ij}, \quad \sum_{j=1}^N T_{ij} = 1. \quad (1)$$

2.1.2. Construction of transition probabilities using gravity models

Transition probabilities between evacuation nodes $v_i, v_j \in \mathcal{V}$ are defined by gravity models based on distance and attractiveness. The weight matrix $W = [w_{ij}]$ is shown below:

$$w_{ij} = \begin{cases} \frac{P_i^\alpha P_j^\beta}{d_{ij}^\gamma}, & \text{if } f_{ij}^{(r_m)} < \rho \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where P_i and P_j are the attractiveness of departure and destination points respectively (evacuee numbers, evacuation shelter capacity, etc.), d_{ij} is distance, $f_{ij}^{(r_m)}$ is a penalty function based on elevation restrictions, and ρ is the cutoff threshold.

By normalizing this weight matrix row-wise, we obtain the transition probability matrix T :

$$T_{ij} = \frac{w_{ij}}{\sum_{k=1}^N w_{ik}}. \quad (3)$$

Since pre-disaster risk assessment typically lacks observed evacuation flow data required for calibration, we adopt baseline parameter settings that minimize model degrees of freedom: $\alpha = \beta = 1$ (linear proportionality to attractiveness) and $\gamma = 2$ (inverse-square distance decay). These are widely used settings in spatial interaction modeling [42,43]. The threshold ρ is a link feasibility condition derived directly from elevation data.

2.1.3. Extraction of equivalence classes using strongly connected components

For the obtained Markov transition matrix T , we construct a directed graph $G = (\mathcal{V}, \mathcal{E})$ and classify the state space into equivalence classes through strongly connected component decomposition. Each equivalence class $\mathcal{E}_k$ satisfies:

$$\forall i, j \in \mathcal{E}_k, \quad \exists n, m \in \mathbb{N} \text{ s.t. } T_{ij}^n > 0, T_{ji}^m > 0 \quad (4)$$

Among these, the equivalence class with the maximum number of nodes $\mathcal{E}_{\max}$ is defined as the main equivalence class and regarded as the primary evacuation area within the network.

2.1.4. Stationary distribution and bias assessment of evacuation destination preferences

For the evacuation shelter node group $S \cap \mathcal{E}_{\max}$ belonging to the main equivalence class, we derive the stationary distribution $\pi = (\pi_1, \dots, \pi_N)$ of the Markov chain. The stationary distribution is defined as the following eigenvector:

$$\pi T = \pi, \quad \sum_{i=1}^N \pi_i = 1. \quad (5)$$

Focusing on evacuation shelter nodes, we define the standard deviation of stationary distribution values as follows:

$$\sigma_\pi^{(s)} = \text{std} \left(\left\{ \pi_i \mid v_i \in S \cap \mathcal{E}_{\max} \right\} \right) \quad (6)$$

This value serves as an indicator showing whether evacuee preferences (= stationary distribution) are excessively concentrated at specific evacuation shelters, quantitatively evaluating the bias and risk concentration of evacuation behavior.

2.2. Geometric coverage and support facility proposals for isolated nodes

We define nodes that do not belong to the main equivalence class (i.e., isolated node groups that lack network connectivity with other evacuation nodes) as $\mathcal{I} \subset \mathcal{V}$. These constitute areas that have lost connection with other evacuation zones or are inaccessible during disasters.

In this study, we regard these isolated nodes as support targets and formulate a geometric minimum covering problem to propose installation locations for new evacuation facilities.

It is essential to clarify the role and interpretation of the geometric coverage model within our framework. The covering circles obtained through this optimization do not represent specific construction sites for evacuation facilities. Rather, they serve as indicators that identify “areas where complementary evacuation support facilities should be considered” and quantify the minimum number of such facilities required to ensure coverage of all isolated nodes.

The actual placement of evacuation facilities involves numerous practical considerations beyond geometric optimization, including land use regulations, building codes and structural requirements, accessibility for diverse populations, terrain suitability, and budget constraints. These factors must be evaluated by policymakers, urban planners, and disaster management practitioners who possess local knowledge and administrative authority. Our method is designed as a decision-support tool that provides quantitative evidence for prioritizing areas that require attention, thereby facilitating more informed and efficient planning processes rather than dictating specific facility locations.

Construction of coverage model

We consider covering all isolated nodes in the set I using variable circles $B(c_j, R_j)$ with maximum radius R (where $R_j \leq R$). Each circle has center c_j where we assume placement of new evacuation shelters (e.g., evacuation towers).

When the evacuation node set covered by each circle $B(c_j, R_j)$ is denoted as $C_j \subseteq I$, the overall coverage condition is formulated as follows:

$$\bigcup_j C_j \supseteq I \quad (7)$$

Here, the objective is to minimize the number of circles used (i.e., the number of newly required evacuation facilities).

Mixed integer linear programming formulation

This problem corresponds to the minimum geometric disk cover problem, which is known to be NP-hard [44]. However, for the problem sizes encountered in practical evacuation planning scenarios, modern optimization solvers can obtain optimal solutions within reasonable computation times. Therefore, we formulate this problem as a Mixed Integer Linear Programming (MILP) model to obtain exact solutions.

Let $\mathcal{G} = \{g_1, g_2, \dots, g_M\}$ be the set of candidate center points generated by discretizing the evaluation area into a regular grid with interval Δ . For each candidate center $g_k \in \mathcal{G}$, we define a binary decision variable:

$$y_k = \begin{cases} 1 & \text{if a covering circle is placed at } g_k \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

For each isolated node $v_i \in I$ and candidate center $g_k \in \mathcal{G}$, we define the coverage indicator:

$$a_{ik} = \begin{cases} 1 & \text{if } \|v_i - g_k\| \leq R \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where $\|v_i - g_k\|$ denotes the Euclidean distance between isolated node v_i and candidate center g_k , and R is the maximum covering radius.

The MILP formulation is then given as follows:

$$\min \sum_{k=1}^M y_k \quad (10)$$

$$\text{s.t. } \sum_{k=1}^M a_{ik} \cdot y_k \geq 1 \quad \forall v_i \in I \quad (11)$$

$$y_k \in \{0, 1\} \quad \forall g_k \in \mathcal{G} \quad (12)$$

The objective function (10) minimizes the total number of covering circles. Constraint (11) ensures that each isolated node is covered by at least one circle. Constraint (12) enforces the binary nature of the decision variables.

To improve computational efficiency, we apply preprocessing to reduce the number of candidate centers by excluding grid points that cannot cover any isolated node within radius R :

$$\mathcal{G}' = \{g_k \in \mathcal{G} \mid \exists v_i \in I : \|v_i - g_k\| \leq R\} \quad (13)$$

This reduced candidate set $\mathcal{G}'$ significantly decreases the problem size while guaranteeing optimal solution quality.

The set of circles obtained through this optimization process is denoted as $D = \{D_1, D_2, \dots\}$, and its cardinality $|D|$ serves as an indicator of the required number of support facilities.

This method enables optimal and quantitative presentation of complementary measures (evacuation shelter installation plans) for evacuation node areas isolated from the main equivalence class.

2.3. Definition of disaster risk indicators

In this study, we define a disaster risk indicator $R(r_m)$ that comprehensively evaluates isolation, number of support facilities, and risk concentration at evacuation shelters for evacuation node network structures obtained under each elevation restriction value r_m .

This indicator comprises the following three components:

- K : Number of equivalence classes (network fragmentation)
- $|D|$: Number of support facilities required through geometric coverage (facility shortage for isolated nodes)
- $\sigma_\pi^{(s)}$: Standard deviation of stationary distributions at evacuation shelter nodes belonging to the main equivalence class (load imbalance)

The stationary distribution π_i represents the long-term proportion of evacuees at each shelter, with $\sigma_\pi^{(s)}$ measuring the degree of concentration: larger values indicate higher overcrowding risk at specific shelters.

Using these components, the disaster risk indicator $R(r_m)$ is formulated as follows:

$$R(r_m) = \lambda \cdot K + \mu \cdot |D| + \nu \cdot \sigma_\pi^{(s)} \quad (14)$$

where λ , μ , and ν are weighting coefficients that adjust the influence of each component, appropriately set considering the scale differences of respective components.

In this study, we calculate this indicator individually for multiple elevation restriction values across a specified range and analyze trends in network structure and disaster risk accompanying elevation changes.

3. Results and discussion

This section reports simulation results using both virtual environments and real data to verify the effectiveness of the proposed disaster risk assessment model. The analysis follows a two-stage validation approach. First, synthetic data experiments are conducted to verify the methodological soundness of our framework, including data independence and model stability under controlled conditions. This step is essential for establishing the generalizability of the method before application to specific real-world contexts. Second, the framework is applied to real-world data from Shizuoka Prefecture to demonstrate practical applicability. This approach ensures that the observed results reflect the inherent properties of the method rather than artifacts of particular datasets.

First, through model implementation using virtual terrain and evacuation node placement, we evaluate structural vulnerabilities based on indicators such as equivalence classes and stationary distributions. Next, through multiple random simulations with varying elevation restrictions, we confirm the consistency and robustness of the risk evaluation indicators. Finally, we construct evacuation networks and conduct risk evaluations using open data from 16 municipalities in Shizuoka Prefecture, Japan, visualizing isolation risks and evacuation difficulties under real topographical conditions. Through this series of analyses, we demonstrate that the proposed disaster risk assessment model is effective for quantifying disaster risks considering topographical and structural factors.

3.1. Model implementation in virtual space and equivalence class analysis

In this section, we evaluate the defined disaster risk indicator (14) using terrain and evacuation node placement generated in virtual space. The terrain is constructed using the following equation by superimposing multiple bivariate normal distribution functions (Gaussian functions):

$$z(x, y) = \sum_{k=1}^{K_z} h_k \cdot \exp\left(-\frac{(x - x_k)^2 + (y - y_k)^2}{2s_k^2}\right) \quad (15)$$

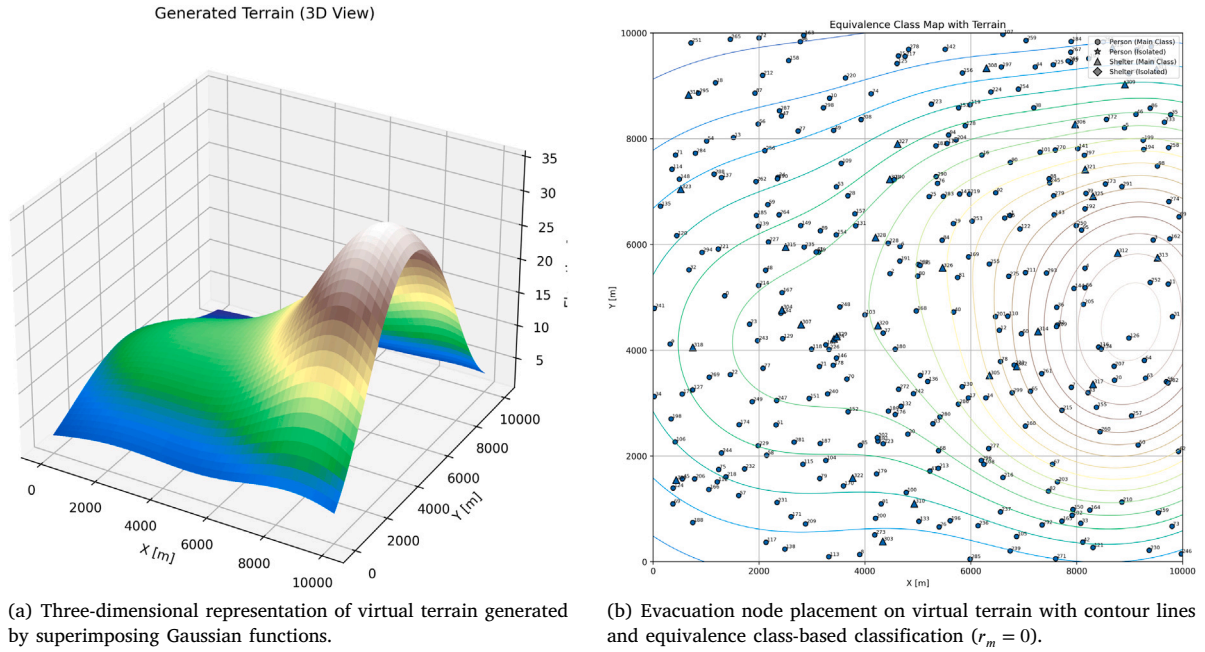


Fig. 1. Three-dimensional representation of virtual terrain and its contour lines with classification results.

where K_z is the number of mountains, (x_k, y_k) represents the position of each peak, h_k is the mountain height, and s_k is the scale parameter representing the spread of the distribution. This generates smooth and continuous elevation distributions similar to real terrain.

Fig. 1(a) shows the three-dimensional representation of terrain generated using this formula. The formation of mountains biased toward the right side (east side) can be confirmed. In this simulation, the domain size was set to $[0, 10000] \times [0, 10000]$ [m] and the elevation range to approximately $[0, \sim 30]$ [m].

On this terrain, we randomly place evacuation targets (N_p persons) and evacuation shelters (N_s locations). Each evacuation node has (x, y, z) coordinates based on elevation, and the evacuation behavior transition probability matrix T is constructed based on the gravity model in Eq. (2), considering weighted distance decay and attractiveness. Subsequently, T is normalized as a probability matrix according to Eq. (3), defining transition probabilities for the discrete-time Markov chain model.

Based on transition probabilities, equivalence classes defined by strong connectivity are extracted according to Eq. (4), and stationary distributions π are derived for evacuation shelter groups belonging to the main equivalence class (Eq. (5)). Furthermore, the standard deviation $\sigma_\pi^{(s)}$ of π for evacuation shelters belonging to the main equivalence class is calculated based on Eq. (6) to evaluate distribution bias.

Fig. 1(b) shows the results of evacuation node placement on virtual terrain and classification by equivalence classes, with contour lines. Evacuation nodes belonging to the main equivalence class are drawn as circles (evacuation targets) and triangles (evacuation shelters), while isolated nodes with no transitions to other evacuation nodes are drawn as stars and diamonds.

3.2. Isolation risk based on elevation restrictions and risk evaluation indicator trends

For each elevation restriction r_m , we evaluate candidate positions for evacuation towers using geometric minimum covering problems for isolated evacuation nodes not belonging to the main equivalence class. Figs. 2(a) and 2(b) show results of covering evacuation nodes outside the main equivalence class with variable-radius covering circles for $r_m = 4$ and $r_m = 15$, respectively. This enables visual demonstration

of evacuation facility placement requirements corresponding to isolated evacuation nodes.

For each elevation restriction r_m , we calculate the disaster risk score $R(r_m)$ (Eq. (14)) using the number of equivalence classes K , the number of covering circles $|D|$, and the standard deviation of stationary distributions for evacuation shelters within the main equivalence class $\sigma_\pi^{(s)}$.

Fig. 3 shows the trends of various evaluation indicators (K , $|D|$, $\sigma_\pi^{(s)}$, $R(r_m)$) accompanying changes in elevation restrictions. As elevation increases, evacuation networks become fragmented, leading to increases in equivalence classes and covering circles, resulting in an upward trend in risk scores.

These results demonstrate that the proposed disaster risk assessment model appropriately reflects the relationship between terrain structure and evacuation networks, enabling quantitative extraction of evacuation nodes with high isolation risk.

3.3. Model stability assessment

In this study, with the objective of evaluating isolation risk based on elevation restrictions, we conducted 10 simulations after randomly generating terrain, evacuees (people), and evacuation shelters (shelters). The simulation in this study uses settings shown in Table 1 for virtual terrain and evacuation node generation. This enables realistic disaster environment simulation while confirming model behavior against elevation changes by varying elevation thresholds in each run.

Fig. 4 shows changes in the number of equivalence classes and standard deviation of stationary distributions in the main equivalence class with respect to elevation from 10 simulation results. The former shows a trend of increasing equivalence class numbers as elevation rises due to evacuation network fragmentation. The latter's increase reflects the tendency for evacuation to concentrate at specific evacuation nodes due to limited accessible evacuation shelters, suggesting evacuation behavior bias.

Note that the number of covering circles in geometric coverage problems also shows a strong correlation ($r \approx 0.97$) with the number of equivalence classes and exhibits similar increasing trends with elevation rise, so visualization is omitted here.

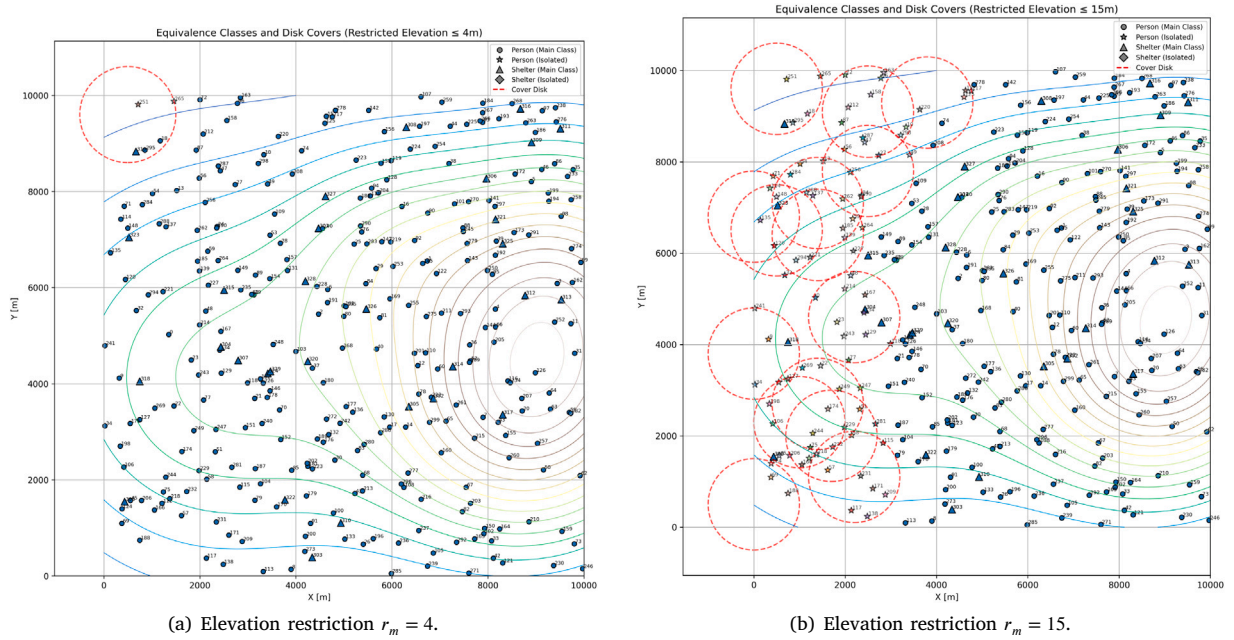


Fig. 2. Equivalence classes and covering circle placement under elevation restrictions.

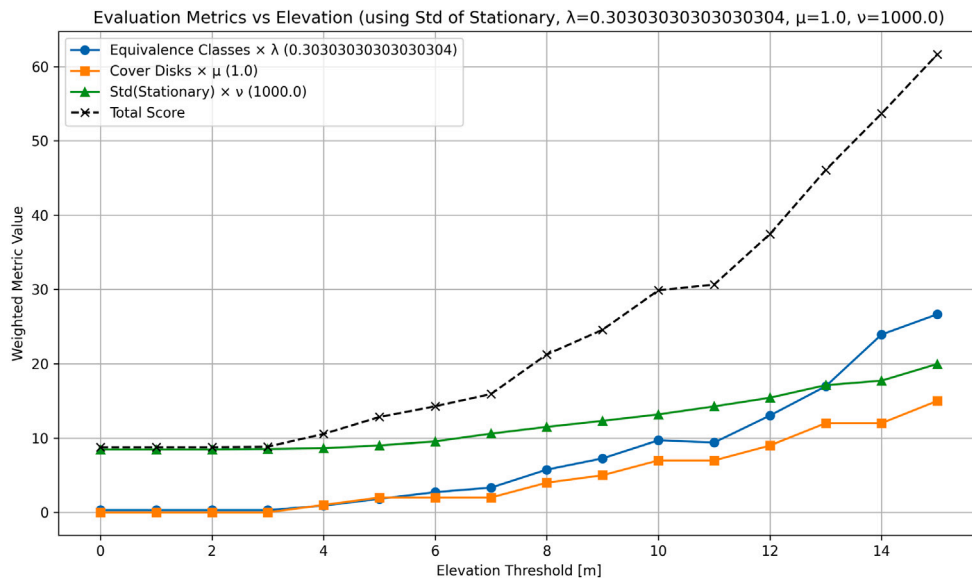


Fig. 3. Trends of evaluation indicators accompanying changes in elevation restriction r_m .

From these results, even across 10 simulations with randomized terrain and evacuation node placement, the proposed disaster risk assessment model demonstrates consistent and coherent trends in each evaluation indicator, confirming high model stability and reproducibility.

3.4. Comparison of optimization methods

To validate the choice of optimization method for the minimum covering circle problem, we conducted a comparative analysis between the Mixed Integer Linear Programming (MILP) approach and the greedy heuristic across 30 simulation runs. This comparison addresses the concern regarding the justification for algorithmic choices and demonstrates that MILP provides optimal solutions within practical computation times for evacuation planning scenarios.

Experimental setup

For each of the 30 simulations with randomly generated terrain and evacuation node placement, we solved the minimum covering circle problem using both methods under identical conditions. The elevation threshold was varied from 0 m to 15 m in 1 m increments, generating a total of 339 cases where isolated nodes existed and required coverage. The MILP formulation (Eqs. (10)–(12)) was solved using the Gurobi optimizer, while the greedy algorithm sequentially selected circles covering the maximum number of uncovered nodes.

Comparative performance

To validate the choice of optimization method for the minimum covering circle problem, we conducted a comparative analysis between the Mixed Integer Linear Programming (MILP) approach and the greedy

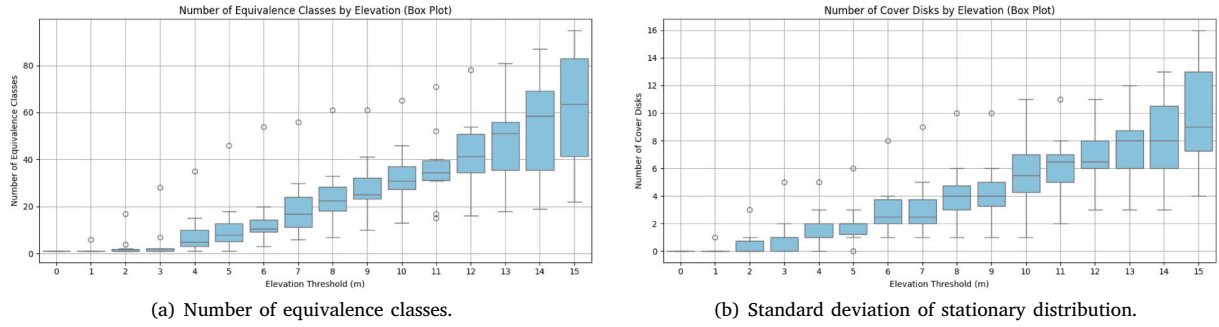


Fig. 4. Distribution of evaluation metrics by elevation threshold.

Table 1

Simulation parameter settings for terrain and node generation.

Parameter	Value/Description
Terrain size	10,000 × 10,000 m ²
Elevation model	Sum of 5 Gaussian hills
Max elevation	~30 m
Number of people	300 (random placement)
Number of shelters	30 (random placement)
Elevation thresholds r_m	0 m to 15 m, step = 1 m
Gravity model parameters	$\alpha = \beta = 1$, $\gamma = 2$
Distance penalty cutoff ρ	Based on lowland fraction matrix
Transition probability	Normalized from weighted gravity model
Transition probability threshold	10^{-3} (for equivalence class analysis)
Equivalence class method	Strongly connected components
Stationary distribution	Calculated from transition matrix
Covering method	MILP-based optimal disk covering for isolated nodes
Evaluation score weights	$\lambda = \frac{100}{330}$, $\mu = 1.0$, $\nu = 1000$

Table 2

Comparison of MILP and greedy approaches for the minimum covering circle problem (30 simulations).

Metric	MILP	Greedy
Solution quality		
Average covering circles	3.92 ± 2.71	4.38 ± 3.26
Computation time		
Average time (ms)	29.82 ± 15.74	11.12 ± 10.06
Maximum time (ms)	84.24	50.68
Performance comparison ($n = 339$)		
MILP better	123 cases (36.3%)	
Equal performance	216 cases (63.7%)	
MILP worse	0 cases (0.0%)	
Statistical tests		
Wilcoxon signed-rank test	$p < 10^{-24}$	
Cohen's d effect size	0.154	

heuristic across 30 simulation runs. Table 2 summarizes the comparative performance across all 339 test cases where isolated nodes existed and required coverage.

Solution quality. Regarding solution quality, MILP required an average of 3.92 ± 2.71 covering circles, while the greedy algorithm required 4.38 ± 3.26 circles. Although MILP produces fewer circles on average, the standard deviations indicate considerable variation across problem instances.

Computation time. In terms of computation time, MILP required an average of 29.82 ± 15.74 ms per case, compared to 11.12 ± 10.06 ms for the greedy approach. The maximum computation time for MILP was 84.24 ms, which remains well within practical limits for emergency planning applications. While greedy is approximately 2.7 times faster, the absolute time differences are negligible in the context of pre-disaster planning.

Performance comparison. Examining the 339 individual test cases, MILP consistently produced solutions equal to or better than the greedy heuristic. Specifically, MILP outperformed greedy in 123 cases (36.3%), achieved identical results in 216 cases (63.7%), and never produced inferior solutions (0 cases, 0.0%). When MILP provided better solutions, the average improvement was 1.28 fewer covering circles, corresponding to an 18.5% reduction. The maximum observed improvement was 4 circles (33.3% reduction), representing meaningful resource savings in disaster preparedness contexts.

Statistical significance. To assess whether the observed difference in solution quality is statistically significant, we applied the Wilcoxon signed-rank test. This non-parametric test compares paired samples (MILP vs. greedy solutions for the same problem instance) without assuming normal distribution, making it appropriate for our data. The test yielded $p < 10^{-24}$, indicating that the superiority of MILP over greedy is statistically significant and not due to random chance. The Cohen's d effect size of 0.154 indicates a small but meaningful practical difference between the two methods.

Analysis by elevation threshold revealed that the advantage of MILP becomes more pronounced for larger problem instances. As elevation thresholds increase from 0 m to 15 m, the number of isolated nodes grows from approximately 12 to 39 on average, creating more complex covering problems. The percentage of cases where MILP outperforms greedy increases from approximately 20% at lower elevations to over 50% at higher elevations (12–14 m), suggesting that optimal solutions provide greater benefits when problem complexity increases.

Justification for MILP adoption

These results justify the adoption of MILP over greedy heuristics for the minimum covering circle problem in evacuation network analysis. The key insights are: (1) MILP guarantees optimal solutions, eliminating the risk of suboptimal facility placement; (2) modern MILP solvers handle municipal-level problem sizes (up to 129 isolated nodes in our experiments) within milliseconds; (3) the relative advantage of MILP increases with problem complexity; and (4) the 18.5% average reduction in required facilities represents meaningful resource savings in disaster preparedness contexts. Based on these findings, we adopt the MILP approach for the subsequent real-data analysis.

3.5. Sensitivity analysis of weight parameters

The weight parameters (λ , μ , ν) in the risk score formulation (Eq. (14)) serve to adjust the relative importance of each component. To examine how the choice of these parameters affects risk assessment outcomes, we conducted a sensitivity analysis using data from 30 independent simulations.

Fig. 5(a) illustrates how each component contributes to the total risk score across different elevation thresholds. At low elevations (0–5 m),

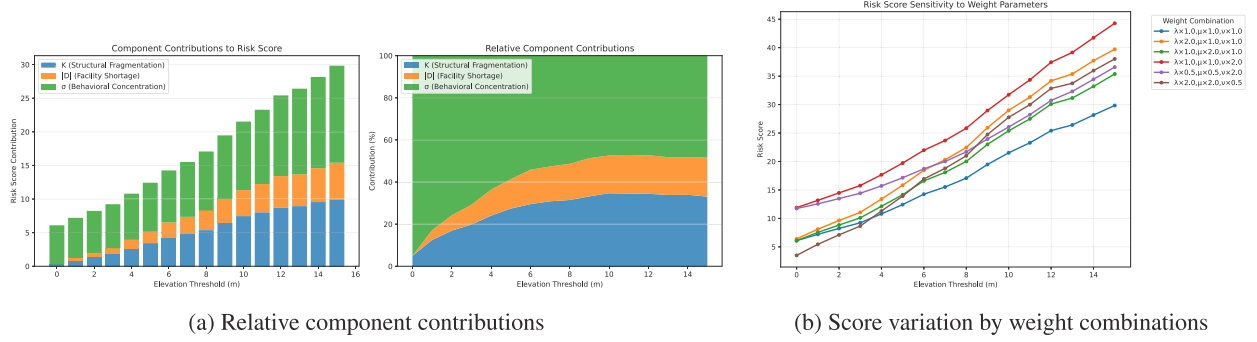


Fig. 5. Sensitivity analysis of weight parameters. (a) Relative component contributions showing the shift from behavioral concentration dominance at low elevations to balanced contributions at higher elevations. (b) Risk score variation under six representative weight combinations, demonstrating consistent monotonic trends with no curve intersections.

the behavioral concentration term $\sigma_x^{(s)}$ dominates, contributing approximately 95% to 59% of the total score, indicating that when the evacuation network remains largely connected, the primary risk factor is the uneven distribution of evacuee preferences toward specific shelters. At higher elevations (10–15 m), the contributions become more balanced: structural fragmentation (K) accounts for approximately 33%–35%, facility shortage ([D]) for 18%–19%, and behavioral concentration for 47%–48%. This shift reflects the increasing importance of network connectivity as elevation restrictions cause fragmentation and isolated areas emerge.

Fig. 5(b) shows risk scores computed using six representative weight combinations. All combinations produce monotonically increasing scores with elevation, confirming that the model consistently captures increasing disaster risk. While absolute values differ — ranging from approximately 6 to 12 at 0 m and from 30 to 44 at 15 m — the relative trends remain consistent. Notably, the curves do not intersect, indicating that relative risk assessment between elevation conditions is preserved regardless of weight selection.

To quantify the robustness of relative risk assessments, we computed Spearman rank correlation coefficients between risk score rankings obtained from all 27 weight combinations (varying each of λ , μ , ν by factors of 0.5, 1.0, and 2.0). All pairwise correlations equal 1.0, indicating perfect rank preservation across all weight combinations. This remarkable stability demonstrates that the framework produces identical relative risk rankings regardless of weight adjustments within the tested range. Consequently, policy makers can adjust weights according to their priorities — emphasizing shelter congestion (ν), infrastructure resilience (λ), or facility adequacy (μ) — without affecting relative risk comparisons between regions or scenarios.

3.6. Risk assessment using real data

3.6.1. Construction of evacuation networks

In this section, to verify the effectiveness of the proposed method, we conducted evacuation network construction and disaster risk assessment using open data targeting 16 municipalities in Shizuoka Prefecture, Japan. Table 3 shows the correspondence between various open data sources used and their respective applications.

First, using census administrative boundary data by town and district, we obtained administrative areas of target municipalities. Additionally, we calculated the centroid coordinates (latitude and longitude) of each area and used them as initial positions for network construction. Next, we integrated and interpolated 10 m mesh elevation data provided by the Geospatial Information Authority of Japan in tile units corresponding to target areas to create elevation arrays. Missing values were supplemented with minimum elevation values, and Gaussian smoothing was applied to ensure continuity of elevation data. The created elevation information was utilized for evaluating evacuation difficulties and as elevation filters during network construction.

For evacuation shelter data, we used “Designated Evacuation Shelter Data” provided by the National Research Institute for Earth Science and Disaster Resilience, extracting evacuation shelters and city halls of relevant municipalities. Each facility was assigned attributes including name, latitude and longitude, priority, and elevation, and configured as evacuation shelter nodes.

Evacuee nodes were allocated based on population statistics by town and district, targeting approximately 1% of the total population according to population ratios. Node positions were randomly placed within each polygon satisfying two conditions: being at least 50 m away from existing evacuation shelters and having an elevation of at least 2 m (to avoid inundation risk). Based on the obtained evacuation shelter nodes and evacuee nodes, we acquired walking networks from OpenStreetMap and calculated elevation profiles for each road edge. Based on this, we removed edges below specified elevations and constructed realistic evacuation networks considering elevation restrictions.

Next, we calculated evacuation route distances (from evacuees to evacuation shelters) on the network and created distance matrices. Based on this distance matrix and evacuation shelter attractiveness (2 for designated emergency evacuation sites, 1 otherwise), we derived Markov transition probability matrices using gravity models. Here, designated emergency evacuation sites are places for temporarily protecting lives during disasters, while designated evacuation shelters are facilities intended for post-disaster residence.

For the obtained Markov transition probabilities, we extracted only transition edges above a threshold (10^{-5}) and constructed directed graphs. Subsequently, we extracted strongly connected components as equivalence classes, defined the largest equivalence class as the “main equivalence class”, and treated other components as isolated areas where evacuation routes were disconnected.

For node groups belonging to the main equivalence class, we calculated stationary distributions of Markov chains using the eigenvector method and evaluated bias in evacuation destination preferences (standard deviation). For isolated nodes not belonging to the main equivalence class, we determined the minimum number of circles covering all nodes within 1 km using integer optimization, defining this as the number of newly required evacuation facilities. The 1 km setting was based on research by [49], which indicated that approximately 60% of elderly people aged 75 and above have difficulty walking distances of 1 km or more to reach evacuation shelters, potentially hindering evacuation. Furthermore, while evacuation sites in Tokyo are generally set within approximately 3 km radius [50], the possibility of encountering unexpected problems such as building collapse or evacuation route blockage increases the urgency of considering nearby alternative evacuation sites, leading to the 1 km setting.

Finally, using the number of equivalence classes, number of required covering circles, and standard deviation of stationary distributions within the main equivalence class, we calculated regional disaster

Table 3
Open data sources and their applications.

Data name	Source	Application
Administrative boundary data (TopoJSON) [45]	NII Geoshape	Regional boundary acquisition and administrative unit division
Elevation data (10 m mesh DEM) [46]	Geospatial Information Authority of Japan “Fundamental Geospatial Data”	Elevation array creation and evacuation-difficult area extraction
Designated evacuation shelter data [47]	Geospatial Information Authority of Japan “Designated Emergency Evacuation Sites and Designated Evacuation Shelters Data”	Evacuation shelter node generation
OpenStreetMap (OSM) data [48]	OpenStreetMap	Road graph generation with elevation filtering

risk scores $R(r_m)$ for each elevation condition. In this study, we set $\lambda = \frac{100}{N_p + N_s}$, $\mu = 1.0$, $\nu = 1000$, where N_p and N_s represent the numbers of evacuees and evacuation shelters, respectively. These values were determined considering the relative importance of each indicator on disaster risk and differences in value scales. First, the number of equivalence classes indicates the number of isolated areas within evacuation networks and is an important structural indicator reflecting the degree of evacuation route disconnection or blockage in regions. Therefore, to provide a certain contribution to the overall risk score, we gave a moderate weight normalized by the total number of evacuation nodes.

The number of covering circles indicates the number of newly required evacuation facilities to resolve isolated areas and is an indicator directly related to practical policy decisions. However, since this number is generally small and has clear units based on physical meaning, we standardized it as $\mu = 1$ to avoid unnecessarily amplifying the overall score.

The standard deviation of stationary distributions in the main equivalence class represents the bias of preferences toward evacuation destinations and encompasses risks of concentration at specific evacuation shelters and capacity overflow. However, since standard deviation values are generally numerically small, we set a large weight of $\nu = 1000$ to prevent the impact on risk scores from being obscured. This appropriately reflects the impact of preference bias on evacuation behavior effectiveness and evacuation shelter operations.

Thus, the weighting parameters are designed based on the characteristics and significance of each component, enabling construction of risk scores that comprehensively reflect multiple perspectives including structural vulnerabilities, facility shortages, and evacuation behavior bias. Note that these weights are examples and can be flexibly adjusted according to regional characteristics and policy priorities.

3.6.2. Calculation and analysis of risk scores by elevation conditions

We conducted 10 random simulations regarding evacuee node placement, constructed evacuation networks from 0 m to 15 m elevation at 1 m intervals, and calculated disaster risk scores for each. The average values are shown in Fig. 6, and the spatial distribution of risk scores under 15 m elevation conditions is shown in Fig. 9.

Fig. 6 shows how average disaster risk scores for each municipality change according to elevation condition changes. Overall, risk scores tend to increase with elevation rise, but clear differences in increase rates are observed among municipalities.

Particularly, municipalities located in coastal lowlands such as Fukuroi City, Iwata City, Yaizu City, and Shimizu and Suruga wards of Shizuoka City show rapid risk score increases from around 10 m elevation, with notable evacuation network disconnections and isolated area formation. In these regions, elevation restrictions cause blockage of major road edges and evacuation routes, manifesting structural vulnerabilities such as evacuation destination bias and facility shortages. Additionally, municipalities such as Kawane Honcho and Mori Town show almost no change in risk scores regardless of elevation. These are located in mountainous regions where residences and evacuation shelters are originally at high elevation, receiving little impact from lowland blockages. In Aoi Ward of Shizuoka City and Kawane Honcho,

despite topographically high elevations, high-isolation-risk areas are scattered due to threshold problems in evacuation behavior transition probabilities, namely distances to evacuation destinations being too far (Fig. 7). This reveals that “being in mountainous areas does not guarantee safety”, and topographical advantages alone cannot sufficiently ensure evacuation network functionality. Particularly when transportation network development and evacuation destination distribution are insufficient, isolated areas may form regardless of elevation.

Based on overall trends, we analyze detailed changes in risk score components for two municipalities showing notable behavior: Shimizu Ward and Yaizu City. Figs. 8(a) and 8(b) show risk score component variations (number of equivalence classes, number of covering circles, standard deviation of stationary distributions) and score fluctuations by elevation conditions for Shimizu Ward and Yaizu City, respectively.

In Shimizu Ward, as shown in Fig. 8(a), risk scores consistently increase with elevation rise. The main factor is the increase in equivalence classes, indicating that elevation restrictions fragment evacuation networks, causing rapid increases in isolated areas. Meanwhile, the standard deviation of stationary distributions remains generally low, indicating limited evacuation destination bias and congestion risks. This suggests that score increases in Shimizu Ward are dominated by structural factors such as decreased accessibility to evacuation shelters and route disconnections, with relatively small impacts from evacuation behavior bias.

In Yaizu City, as seen in Fig. 8(b), scores temporarily decrease significantly around 13 m elevation. This behavior corresponds to rapid decreases in standard deviation of stationary distributions compared to other indicators. This is considered a result of some evacuation shelters being blocked by elevation filters, causing temporary preference homogenization. In other words, partial evacuation network fragmentation conversely alleviated preference concentration, resulting in reduced bias and lower scores.

By examining component-wise variations, we understand that risk score changes are sensitively affected not only by simple network disconnections or evacuation shelter shortages but also by bias in evacuee preference distributions (= congestion risk). Particularly, cases like Yaizu City, where network structure changes have unexpected positive and negative impacts on evacuation behavior distribution, suggest the importance of planned evacuation guidance.

Fig. 9 shows the spatial distribution of disaster risk scores when applying 15 m elevation restrictions. High-risk areas (red tones) are concentrated in low-elevation coastal areas including Suruga Ward, Yaizu City, Fukuroi City, and Iwata City, geographically visualizing evacuation difficulties due to network fragmentation. In contrast, inland areas such as Mori Town, Shimada City, and Fujieda City show relatively low risk scores with small impacts from elevation restrictions.

These results demonstrate that the impact of elevation conditions on regional evacuation networks strongly depends on topographical factors and urban structure. Particularly, municipalities in low-elevation coastal areas require measures such as evacuation facility relocation and route diversification since networks are rapidly fragmented when elevation thresholds are exceeded. Meanwhile, inland mountainous areas maintain stable risks with evacuation routes easily secured under

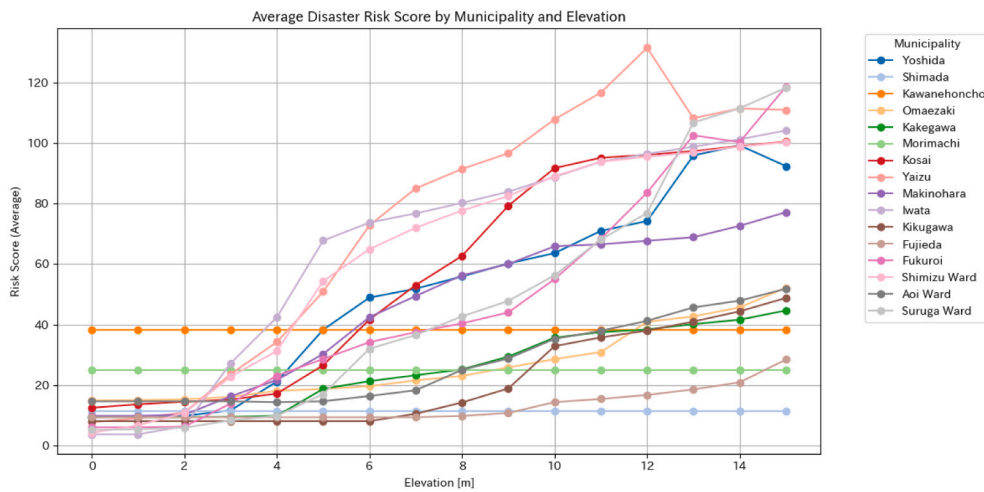


Fig. 6. Trends of evaluation indicators accompanying changes in elevation restriction r_m .

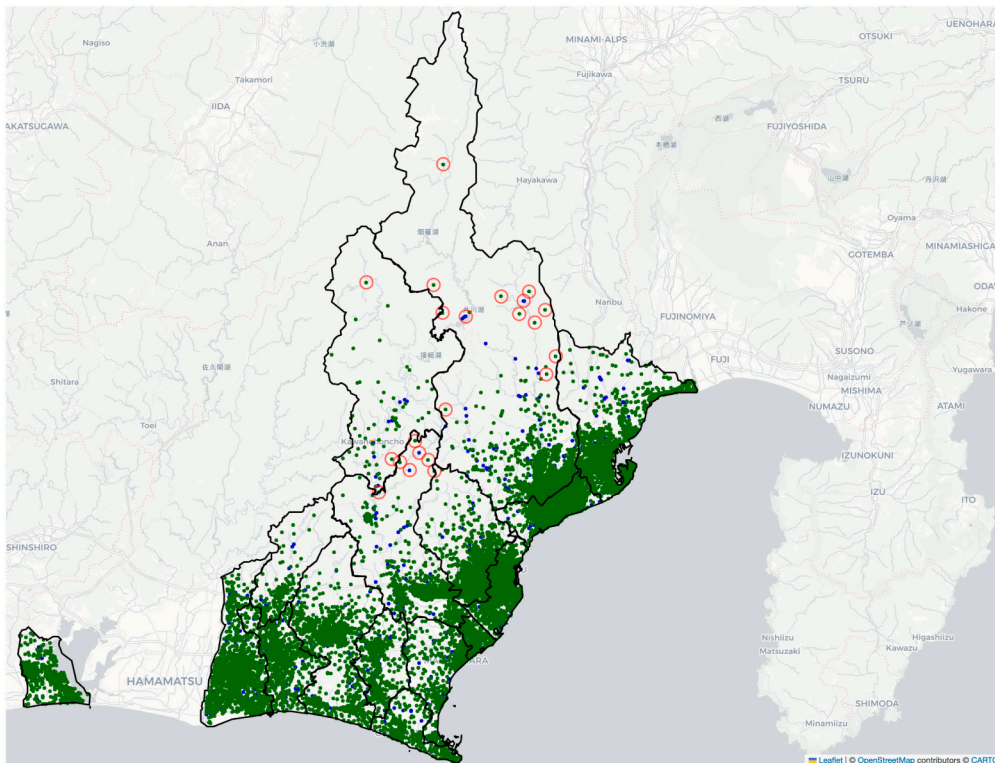


Fig. 7. Coverage status of evacuation-difficult areas under 0 m elevation restriction (red circles: covering circles, blue points: evacuation shelters, green points: evacuees).

current geographical structures. This method serves as an effective evaluation indicator for quantitatively understanding evacuation difficulties based on topographical differences and prioritizing regional response measures. Additionally, by comparing evacuation difficulties among regions, effective decision support for evacuation shelter development and regional planning becomes possible.

4. Conclusion

This study proposed a novel disaster risk assessment method combining Markov chains and geometric coverage theory to quantitatively evaluate the structural robustness and isolation risks of evacuation networks during disasters. Specifically, we constructed entire evacuation

networks as discrete-time Markov chains using probabilistic transitions between evacuation nodes estimated through gravity models, performed equivalence class classification using strongly connected components, stationary distribution analysis for main equivalence classes, and complementary design through covering circle placement for isolated nodes.

Through 10 simulations on virtual terrain, the proposed disaster risk assessment model demonstrated consistent and coherent variations in evaluation indicators in response to elevation condition changes, confirming model stability and reproducibility. Furthermore, empirical analysis using real data from 16 municipalities in Shizuoka Prefecture, Japan, successfully visualized structural vulnerabilities including evacuation network fragmentation due to topographical constraints, manifestation of isolated areas, and bias in evacuation destination

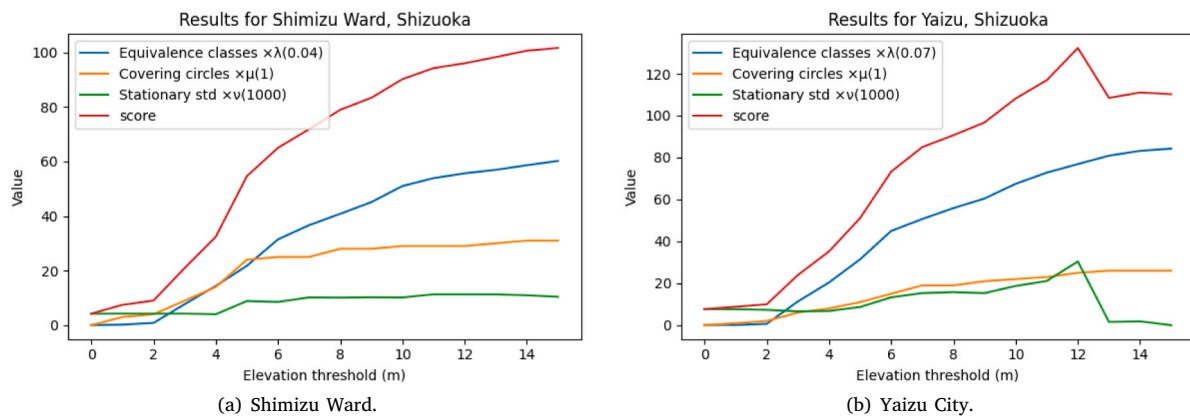


Fig. 8. Changes in risk score components.

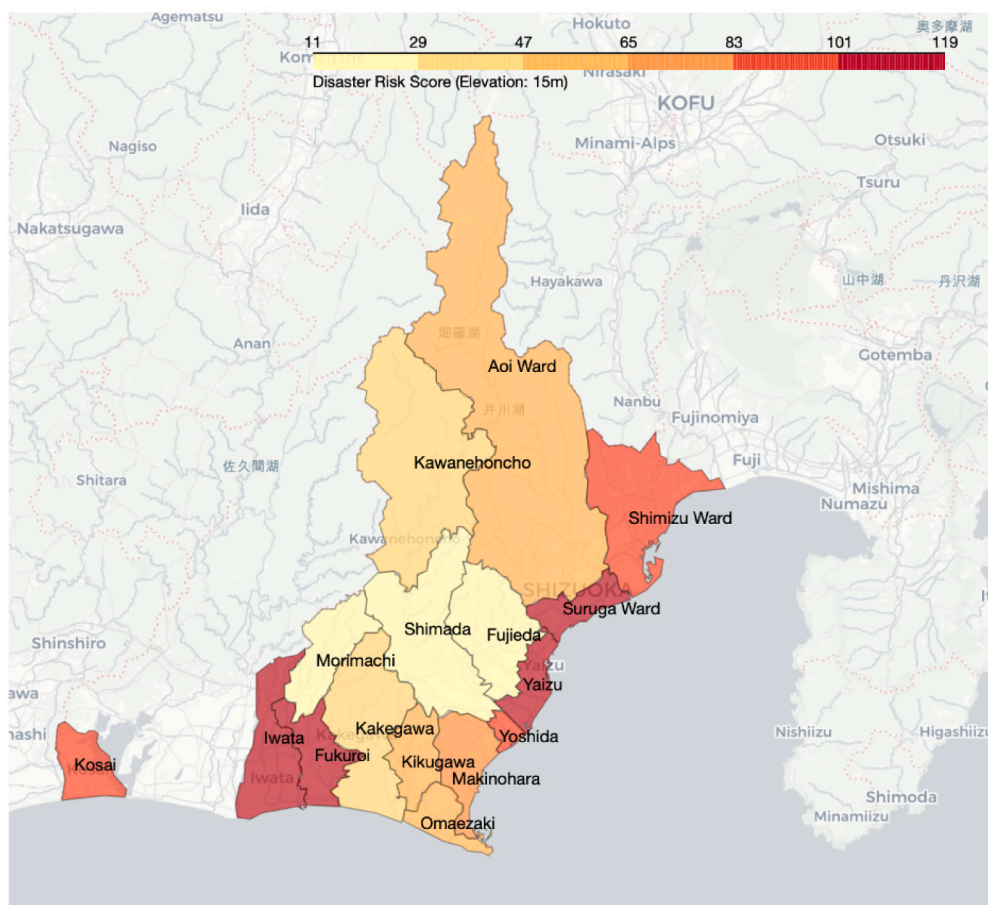


Fig. 9. Spatial distribution of disaster risk scores under 15 m elevation restriction.

preferences, demonstrating utility for regional disaster risk comparison and policy intervention prioritization.

This method can serve as a tool supporting more practical and flexible evacuation support planning that goes beyond conventional simple evacuation shelter placement evaluations based on distance and time by comprehensively considering geographical and structural factors. Unlike optimization-driven approaches that prescribe specific facility placements or evacuation routes, this method provides structure-centered diagnostic insights that reveal inherent vulnerabilities in existing networks, enabling planners to systematically identify where

interventions are most needed before determining specific countermeasures.

From the perspective of practical application, this method functions effectively as a pre-disaster planning tool. By conducting risk assessments corresponding to multiple disaster scenarios during normal times and integrating the results with hazard maps and evacuation plans, it is possible to establish a foundation for rapid decision-making when disasters occur. Future extensions are expected to include dynamic evaluation through integration with real-time inundation data and visualization systems through linkage with GIS platforms.

Future research directions include several important extensions. First, while this study used elevation restrictions as proxy indicators for disaster impacts, integrating detailed hazard map data for floods, landslides, tsunamis, and other disasters would enable more comprehensive and precise risk assessments. In Japan, high-precision hazard maps developed by the Ministry of Land, Infrastructure, Transport and Tourism are available nationwide, and method validation is expected in various global regions where similar data are available. Second, addressing compound disaster occurrences and time-dynamic network changes accompanying disaster progression. Third, integration with individual support plans and diverse mobility modes (vehicle evacuation, public transportation). Fourth, extension to multi-objective optimization through consistency verification with actual human flow data and evacuation behavior surveys. It should be noted that validation against real evacuation behavior is beyond the scope of this study and remains an important direction for future work. Through these developments, this method is expected to contribute to improving the effectiveness of regional disaster prevention planning and strengthening disaster resilience.

CRediT authorship contribution statement

Shinya Mizuno: Writing – review & editing, Supervision, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Haruka Ohba:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used generative AI (ChatGPT, OpenAI) to improve the clarity and readability of the English writing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data are publicly available from cited open sources. Code: <https://github.com/smzn/markov-evacuation-network-risk-assessment>.

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